

Madrid Data Science – Technical Assessment

Objective

We aim to evaluate the candidate's ability to analyze data and assess the effectiveness of two different pricing strategies (A and B). The candidate is expected to provide reproducible code and a non-technical presentation of the findings.

Dataset Description

The provided dataset contains flight booking information with the following columns:

- `dep_date`: Date of flight departure.
- `model`: Pricing strategy used (either A or B).
- `inv_leg_id`: Unique flight identifier.
- `booking_ts`: Timestamp of when the booking occurred.
- `passenger_id`: Unique passenger identifier.
- `fare_eur`: Amount paid by the passenger in EUR.
- `is_active`: Boolean flag indicating whether the booking is still active (1) or canceled/moved (0).

For the third exercise we do not have the `model` column and there is a column called `route_group` that is a label that describes an aggregation of flights.

Task Description

The assessment consists of three parts:

- **Part 1: Flight Clustering**
 - Identify meaningful clusters of flights.
 - Justify the choice of clustering methodology and provide insights into the characteristics of the clusters. For that purpose, you can ignore `model` column as it will be used only for Part 2.
- **Part 2: Pricing Model Evaluation**
 - Assess the performance of pricing strategies A and B.
 - Define metrics to compare the models.
 - Provide a clear recommendation on which strategy is more effective based on the data.
- **Part 3: Classification problem**
 - You are provided with a second dataset that includes a column called `route_group`. Build a model capable of predicting the category for each flight.
 - Clearly state the metrics and methodology you use to assess the model's performance.
 - Describe how this model could be implemented and maintained in production.

Deliverables

A well-documented script with clear instructions on how to execute the analysis on a local machine.

Presentation (~30 minutes)

A summary of the analysis, findings, and recommendations.

The presentation should be accessible to a non-technical audience, focusing on key business insights rather than technical details.

Timeline

- The task should take approximately 2-3 working days.
- The candidate has 10 days to submit the complete solution.

We look forward to evaluating your approach and insights!